

NOTE ON VARIABLE ROLE-REVERSALS

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ABSTRACT. In [2], the author uses variable role-reversals only when $\theta < \frac{45}{89} - \varepsilon$. In this note, we shall discuss how to use variable role-reversals when $\frac{45}{89} - \varepsilon \leq \theta < \frac{11}{20} - \varepsilon$.

In a recent paper [2], the author studied the average distribution of majorants and minorants of $\mathbb{1}_p(x)$ in arithmetic progressions to moduli larger than $\sqrt{x}$. During the constructing process of those majorants and minorants, the author applied the *variable role-reversal*, which is also a very powerful technique in many other applications of Harman's sieve. For the sake of simplicity, the author only uses role-reversals when $\theta < \frac{45}{89} - \varepsilon$. Here, we generalize the application of role-reversals in [2] to a wider range $\theta < \frac{11}{20} - \varepsilon$. We mention that further extension to $\theta < \frac{4}{7} - \varepsilon$ may still be possible.

From here we suppose that $\frac{1}{2} \leq \theta < \frac{11}{20} - \varepsilon$. Almost all of the notations and definitions in this note are same as those in [2]. We only change the definitions of κ and τ in order to state our Lemma 0.2 and Lemma 0.3 below simpler. Note that κ here equals to κ in [2] when $\theta \leq \frac{7}{13} - \varepsilon$ and becomes κ' in [2] when $\theta > \frac{7}{13} - \varepsilon$. The definitions of some sets below are changed correspondingly.

$$\kappa = \kappa(\theta) = \begin{cases} \frac{5-8\theta}{6} - \varepsilon, & \theta \leq \frac{17}{32} - \varepsilon, \\ \frac{5-8\theta}{12} - 3\varepsilon, & \frac{17}{32} - \varepsilon < \theta \leq \frac{7}{13} - \varepsilon, \\ \frac{11-20\theta}{6} - 2\varepsilon, & \frac{7}{13} - \varepsilon < \theta \leq \frac{11}{20} - \varepsilon, \end{cases}$$

$$\tau = \tau(\theta) = \begin{cases} \frac{3(1-\theta)}{5} - \varepsilon, & \theta \leq \frac{11}{21}, \\ \frac{2}{7} - \varepsilon, & \frac{11}{21} < \theta \leq \frac{6}{11} - \varepsilon, \\ \frac{5-6\theta}{7} - \varepsilon, & \frac{6}{11} - \varepsilon < \theta. \end{cases}$$

$$\mathbf{S} = \mathbf{S}(\theta) = \{(s, t) : s \leq 1 - \theta - \varepsilon, s + 2t \leq 2 - 2\theta - \varepsilon, s + 4t \leq 2 - \theta - \varepsilon\},$$

$$\mathbf{S}_j = \mathbf{S}_j(\theta) = \{\alpha_j : \alpha_j \text{ partitions exactly into } \mathbf{S}\},$$

$$\mathbf{A}_j = \mathbf{A}_j(\theta) = \{\alpha_j : (\log \log x)^{-3} \leq \alpha_j \leq \dots \leq \alpha_1 < \tau, \alpha_1 + \dots + \alpha_j \leq 1\},$$

$$\mathbf{A}'_j = \mathbf{A}'_j(\theta) = \left\{ \alpha_j : (\log \log x)^{-3} \leq \alpha_j \leq \dots \leq \alpha_1 < \frac{3}{7} + \varepsilon, \alpha_1 + \dots + \alpha_j \leq 1 \right\},$$

$$\mathbf{U}_j = \mathbf{U}_j(\theta) = \{\alpha_j : \alpha_j \in \mathbf{A}_j, (\alpha_1, \dots, \alpha_j, 2\theta - 1 + \varepsilon) \in \mathbf{S}_{j+1}\},$$

$$\mathbf{U}'_j = \mathbf{U}'_j(\theta) = \{\alpha_j : \alpha_j \in \mathbf{A}'_j, (\alpha_1, \dots, \alpha_j, 2\theta - 1 + \varepsilon) \in \mathbf{S}_{j+1}\}.$$

The next three lemmas come directly from [2].

Lemma 0.1. Let $\frac{1}{2} \leq \theta \leq \frac{45}{89} - \varepsilon$, $m_1 \sim M_1$ and $m_2 \sim M_2$. Then

$$\sum_{\left(\frac{\log M_1}{\log x}, \frac{\log M_2}{\log x}\right) \in \mathbf{S}} a_{1,m_1} a_{2,m_2} S(\mathcal{A}_{m_1 m_2}^q, x^\kappa)$$

has an asymptotic formula.

Proof. This lemma is [[2], Lemma 2.14]. □

Lemma 0.2. Let $\frac{1}{2} \leq \theta \leq \frac{17}{32} - \varepsilon$. Then

$$\sum_{\alpha_j \in \mathbf{U}'_j} S(\mathcal{A}_{p_1 \dots p_j}^q, x^\kappa)$$

has an asymptotic formula.

Proof. This lemma can be proved in the same way of [[2], Lemma 2.13]. $\square$

Lemma 0.3. Let $\frac{1}{2} \leq \theta \leq \frac{11}{20} - \varepsilon$. Then

$$\sum_{\alpha_j \in U_j} S\left(\mathcal{A}_{p_1 \dots p_j}^q, x^\kappa\right)$$

has an asymptotic formula.

Proof. This lemma can be proved in the same way of [[2], Lemma 2.13]. $\square$

Next we shall briefly introduce the use of role-reversals in this problem. Suppose that we want to give a nontrivial estimate for the sum

$$\sum_{\alpha_j} S\left(\mathcal{A}_{p_1 \dots p_j}^q, p_j\right). \quad (1)$$

Assume that the sum (1) has a “right sign” (that is, $-$ in the construction of a majorant or $+$ in the construction of a minorant) and satisfies $\alpha_j \in \mathbf{S}_j$ (or $\mathbf{U}'_j$ or $\mathbf{U}_j$, depending on the value of θ). A direct application of Buchstab’s identity gives

$$\sum_{\alpha_j} S\left(\mathcal{A}_{p_1 \dots p_j}^q, p_j\right) = \sum_{\alpha_j} S\left(\mathcal{A}_{p_1 \dots p_j}^q, x^\kappa\right) - \sum_{\substack{\alpha_j \\ \kappa \leq \alpha_{j+1} < \min(\alpha_j, \frac{1}{2}(1-\alpha_1-\dots-\alpha_j))}} S\left(\mathcal{A}_{p_1 \dots p_j p_{j+1}}^q, p_{j+1}\right). \quad (2)$$

After applying Buchstab’s identity once, sometimes we cannot use it again since $(\alpha_1, \dots, \alpha_j, \alpha_j) \notin \mathbf{S}_{j+1}$ (or $\mathbf{U}'_{j+1}$ or $\mathbf{U}_{j+1}$). In this case, we cannot give an asymptotic formula for the sum (with a “wrong sign”)

$$\sum_{\substack{\alpha_j \\ \kappa \leq \alpha_{j+1} < \min(\alpha_j, \frac{1}{2}(1-\alpha_1-\dots-\alpha_j))}} S\left(\mathcal{A}_{p_1 \dots p_j p_{j+1}}^q, x^\kappa\right) \quad (3)$$

after the second Buchstab iteration. The idea of a role-reversal is to apply the second Buchstab iteration on a different variable, usually the largest explicit prime variable (denoted by p_1):

$$(p_1 p_2 \dots p_{j+1})\beta \Rightarrow (\beta p_2 \dots p_{j+1})p_1.$$

Reversing the roles of β and p_1 gives

$$\begin{aligned} & \sum_{\substack{\alpha_j \\ \kappa \leq \alpha_{j+1} < \min(\alpha_j, \frac{1}{2}(1-\alpha_1-\dots-\alpha_j))}} S\left(\mathcal{A}_{p_1 \dots p_j p_{j+1}}^q, p_{j+1}\right) \\ &= \sum_{\substack{\alpha_j \\ \kappa \leq \alpha_{j+1} < \min(\alpha_j, \frac{1}{2}(1-\alpha_1-\dots-\alpha_j))}} S\left(\mathcal{A}_{\beta p_2 \dots p_j p_{j+1}}^q, \left(\frac{2x}{\beta p_2 \dots p_{j+1}}\right)^{\frac{1}{2}}\right). \end{aligned} \quad (4)$$

Then, using the second Buchstab’s identity (actually on the variable p_1 in (3)), we have

$$\begin{aligned} & \sum_{\alpha_{j+1}} S\left(\mathcal{A}_{\beta p_2 \dots p_j p_{j+1}}^q, \left(\frac{2x}{\beta p_2 \dots p_{j+1}}\right)^{\frac{1}{2}}\right) \\ &= \sum_{\alpha_{j+1}} S\left(\mathcal{A}_{\beta p_2 \dots p_j p_{j+1}}^q, x^\kappa\right) - \sum_{\substack{\alpha_{j+1} \\ \kappa \leq \alpha_{j+2} < \frac{1}{2}\alpha_1}} S\left(\mathcal{A}_{\beta p_2 \dots p_j p_{j+1} p_{j+2}}^q, p_{j+2}\right). \end{aligned} \quad (5)$$

Note that $\beta \asymp x^{1-\alpha_1-\dots-\alpha_{j+1}}$. Now we only need the condition $(1-\alpha_1-\dots-\alpha_j, \alpha_2, \dots, \alpha_j) \in \mathbf{S}_j$ (or $\mathbf{U}'_j$ or $\mathbf{U}_j$) instead of $(\alpha_1, \dots, \alpha_j, \alpha_j) \in \mathbf{S}_{j+1}$ (or $\mathbf{U}'_{j+1}$ or $\mathbf{U}_{j+1}$). We can then discard parts of the last sum on the right-hand side of (5) with $(1-\alpha_1-\dots-\alpha_{j+1}, \alpha_2, \dots, \alpha_{j+2}) \notin \mathbf{G}_{j+2}$. More role-reversals are also possible to apply on other variables (e.g. $p_2, p_{j+2}, \dots$) of this sum.

When $\theta \leq \frac{45}{89} - \varepsilon$, $\alpha_j \in \mathbf{S}_j$ and $(1-\alpha_1-\dots-\alpha_j, \alpha_2, \dots, \alpha_j) \in \mathbf{S}_j$, by the definition of $\mathbf{S}_j$, we can give asymptotic formulas for the first sums on the right-hand side of (2) and (5) using Lemma 0.1 directly. In [2] the author indeed used Lemma 0.1 to give an asymptotic formula for the first sum on the right-hand

side of (5). When $\theta > \frac{45}{89} - \varepsilon$, however, the available results (Lemmas 0.2–0.3) require a descending order between prime variables in the definitions of $\mathbf{A}_j$ and $\mathbf{A}'_j$. Thus, at least a splitting argument on the (possibly non-prime) variable β is necessary.

Since we have $(\beta, P(p_{j+1})) = 1$ and $\alpha_1, \alpha_2, \dots, \alpha_{j+1} \geq \kappa$, any prime factor of β is $\geq x^\kappa$, and $\frac{\log \beta}{\log x} = 1 - \alpha_1 - \dots - \alpha_{j+1} \leq 1 - (j+1)\kappa$. Hence, the number of prime factors of β is no more than

$$\left\lceil \frac{1 - (j+1)\kappa}{\kappa} \right\rceil < \infty. \quad (6)$$

Now, we can split β into a product of finitely many prime variables. Write $\beta = p'_1 p'_2 \dots p'_l$, $p'_i \asymp x^{\alpha'_i}$ and $\alpha'_1 \geq \alpha'_2 \geq \dots \geq \alpha'_l \geq \alpha_{j+1} \geq \kappa$. Note that we have $1 - \alpha_1 - \dots - \alpha_{j+1} = \alpha'_1 + \dots + \alpha'_l$.

Case 1. $\frac{45}{89} - \varepsilon < \theta \leq \frac{17}{32} - \varepsilon$. In order to give an asymptotic formula for the first sum on the right-hand side of (2), we need $\alpha_j \in \mathbf{U}'_j$, and thus $\alpha_1 < \frac{3}{7} + \varepsilon$. We also need to give an asymptotic formula for the first sum on the right-hand side of (5). Assume first that $\alpha'_1 < \alpha_1$. Since

$$\alpha_1 > \alpha_2 > \dots > \alpha_{j+1} \geq \kappa \quad \text{and} \quad \alpha_1 > \alpha'_1 \geq \alpha'_2 \geq \dots \geq \alpha'_l \geq \alpha_{j+1} \geq \kappa,$$

we know that the $j+l$ variables $\alpha_2, \dots, \alpha_{j+1}, \alpha'_1, \dots, \alpha'_l$ can be arranged into a descending order (denoted by α'_{j+l}) with the largest variable ($= \max(\alpha_2, \alpha'_1)$) less than α_1 . Hence, we have $\alpha'_{j+l} \in \mathbf{A}'_{j+l}$. Now, we only need the condition $(\alpha'_{j+l}, 2\theta - 1 + \varepsilon) \in \mathbf{S}_{j+l+1}$, which can be satisfied if we have

$$(\alpha_2, \dots, \alpha_j, \alpha_{j+1}, 1 - \alpha_1 - \dots - \alpha_{j+1}, 2\theta - 1 + \varepsilon) \in \mathbf{S}_{j+2}.$$

Again, this can be satisfied if we have

$$(\alpha_2, \dots, \alpha_j, 1 - \alpha_1 - \dots - \alpha_j, 2\theta - 1 + \varepsilon) \in \mathbf{S}_{j+1}.$$

Assume that $\alpha'_1 \geq \alpha_1$. Then we have $1 - \alpha_1 - \dots - \alpha_{j+1} = \alpha'_1 + \dots + \alpha'_l > \alpha'_1 \geq \alpha_1$. Now, if we have

$$(\alpha_2, \dots, \alpha_j, \alpha_{j+1}, 1 - \alpha_1 - \dots - \alpha_{j+1}, 2\theta - 1 + \varepsilon) \in \mathbf{S}_{j+2},$$

we must have

$$(\alpha_1, \alpha_2, \dots, \alpha_j, \alpha_{j+1}, 2\theta - 1 + \varepsilon) \in \mathbf{S}_{j+2},$$

and we can give an asymptotic formula for the sum (3). Thus, a role-reversal is not necessary to apply in this case, and we can just perform another direct Buchstab iteration on the last sum on the right-hand side of (2) to get the sums (3) and

$$\sum_{\substack{\alpha_j \\ \kappa \leq \alpha_{j+1} < \min(\alpha_j, \frac{1}{2}(1 - \alpha_1 - \dots - \alpha_j)) \\ \kappa \leq \alpha_{j+2} < \min(\alpha_{j+1}, \frac{1}{2}(1 - \alpha_1 - \dots - \alpha_j - \alpha_{j+1}))}} S\left(\mathcal{A}_{p_1 \dots p_j p_{j+1} p_{j+2}}^q, p_{j+2}\right). \quad (7)$$

We can then discard parts of (7) with $\alpha_{j+2} \notin \mathbf{G}_{j+2}$.

Case 2. $\frac{17}{32} - \varepsilon < \theta < \frac{11}{20} - \varepsilon$. The discussions are very similar to those in **Case 1**. The only difference is that we need $\alpha_j \in \mathbf{U}_j$ instead of $\alpha_j \in \mathbf{U}'_j$ (Lemma 0.2 is not available here), so that $\alpha_1 < \tau$.

In other similar cases where $2\theta - 1 + \varepsilon$ is replaced by $2\theta_1 + \theta_2 - 1 + \varepsilon$ or ε (see [2]), and when performing further role-reversals on a sum that a role-reversal has already been performed on it (which means that more almost-prime variables become explicit, see [[1], (34)] for example), the required conditions are similar.

As an application, we get the following better bounds for $C_0^*(\theta)$ using the **first method** in [2] by applying role-reversals in the extended range of θ . Many other bounds in [2] can be improved using a similar approach.

θ	$C_0^*(\theta)$ (Old)	$C_0^*(\theta)$ (New)	θ	$C_0^*(\theta)$ (Old)	$C_0^*(\theta)$ (New)
0.501	0.8636	0.8636	0.517	0.4455	0.4936
0.502	0.8455	0.8455	0.518	0.4086	0.4624
0.503	0.8298	0.8298	0.519	0.3744	0.4261
0.504	0.8135	0.8135	0.520	0.3363	0.3890
0.505	0.7972	0.7972	0.521	0.2973	0.3510
0.506	0.7299	0.7730	0.522	0.2546	0.3113
0.507	0.7082	0.7505	0.523	0.2148	0.2714
0.508	0.6894	0.7334	0.524	0.1706	0.2277
0.509	0.6606	0.7045	0.525	0.1218	0.1785
0.510	0.6369	0.6808	0.526	0.0778	0.1324
0.511	0.6128	0.6551	0.527	0.0267	0.0779
0.512	0.5907	0.6373	0.528	0	0.0256
0.513	0.5567	0.6033	0.529	0	0
0.514	0.5332	0.5796	0.530	0	0
0.515	0.5040	0.5496	0.531	0	0
0.516	0.4762	0.5245			

Table 1: Lower Bounds for $C_0^*(\theta)$ (First Method, $\frac{1}{2} < \theta \leq \frac{17}{32} - \varepsilon$)

REFERENCES

- [1] R. Li. On the largest prime factor of integers in short intervals III. *arXiv e-prints*, page arXiv:2601.00910v1, 2026.
- [2] R. Li. Primes in arithmetic progressions to large moduli and refinements of Harman's sieve. *arXiv e-prints*, page arXiv:2602.20917v6, 2026.

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